RESEARCH AND INVESTMENT SCREENING

ZeroWatt Wind

Evidence-backed feasibility and business report

Wind-driven mechanical ventilation and supplemental pond aeration in Bangladesh

Decision: fund a gated validation pilot; commercial rollout is not yet supported.

Prepared for renewable-energy judges, investors and university evaluators.

Research cut-off: 22 October 2026 (Asia/Dhaka).

Baseline: supplied project submission, poster and audited repository source.

Reading the evidence. [S] is an attributed observation or published value; [D] is a reproducible calculation; [A] is an explicit scenario assumption. A source that describes a prototype does not validate its output. Low, base and high allowances are not supplier quotations. Values marked NOT FOUND remain unresolved. All figures are vector drawings embedded in this single file.

Executive summary

Verdict. The printed prototype demonstrates a mechanical idea, while the poster expressly leaves airflow, starting wind speed, torque, drivetrain loss and oxygen transfer unmeasured. [35, 36] The business should purchase evidence before purchasing scale. A limited, supervised supplemental-use pilot is defensible; a claim of guaranteed clinical ventilation or fish-survival protection is not.

Base case headline	Result	Basis
Project NPV, nominal BDT	−11,181,078	[D] monthly FCF, assumed required return
Project NPV, USD equivalent	−90,425	[D] dated conversion
Annual effective project IRR	−47.7%	[D] negative; unique bracketed root
Simple / discounted payback	Not reached within model horizon	[D] cumulative cash flow remains negative
Initial investment / peak cumulative funding need	2,240,000 / 21,769,906	[D] no grant or exit-value rescue
Basic A / pond B / combined A+B ex-works BOM	40,000 / 43,200 / 83,200	[D] from [A] production allowances

The base screening assumes a site mean wind of 3.5 [A] m/s. It predicts full additional-flow performance for 29.2% [D] of time after mechanical availability, distinct from an energy displacement fraction of 47.1% [D]. These probabilities come from an assumed distribution, not a Bangladesh rooftop measurement.

The NPV discount rate is 18% [A], tested alongside 12% [A] and 25% [A]. The historical Bangladesh Bank policy benchmark was 10% [S]; the project risk premium is an analyst assumption. [37]

Commercial tariffs, willingness to pay, adoption, component quotations and durability remain weakly evidenced. The optimistic scenario is conditional and profitable; it is not a probability-weighted forecast. Currency: 123.65 [S] BDT/USD, Bangladesh Bank weighted average dated 2026-10-19, accessed 2026-10-22. [25]

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1 Problem sizing and the addressable need

Poor indoor air and pond oxygen matter, but their national consequences must not be attributed wholesale to this device. World Bank estimates combine outdoor and household pollution, unsafe water, sanitation and lead; they are not a ventilation-only loss estimate. [46]

Evidence / boundary	Value	Basis
National pond area	424,168 ha	[S] DoF, reporting period 2023-24 [15]
Number of ponds	2,543,869	[S] DoF table on culture intensity; not paying customers [15]
Annual pond fish production	2,368,741 tonnes	[S] DoF [15]
National combined pollution burden	Over 272,000 premature deaths; 5,200,000,000 illness days/year	[S] Context only [46]
Household and outdoor air pollution health cost	8.32% of GDP (2019)	[S] Historical national burden [46]

The project repeats a claim linking 500 [S] ppm additional CO2 with a 15 [S]% cognitive-performance reduction. The institutional description and primary-study record concern specific observational test endpoints. A general causal productivity loss at this exact percentage was NOT VERIFIED and is excluded from revenue, health savings and NPV. [35, 29, 13]

FAO describes oxygen depletion and reduced growth in Bangladesh ponds, including warm calm nights and cloudy monsoon conditions. A Bangladesh heatwave report corroborates the mechanism but does not quantify national losses attributable only to low DO. These sources do not establish this prototype prevents fish kills. [18, 30]

1.1 A site-level loss exposure, not an avoided-loss promise

An illustrative 1,000 [A] kg stock at 200 [A] BDT/kg and a 10% [A] event loss exposes 20,000 [D] BDT. Low/high loss fractions are 2% [A] / 30% [A]. The event probability and avoidable fraction are NOT FOUND; no expected-value benefit is booked. Time spent manually splashing, fetching fuel or servicing the unit is likewise NOT FOUND; travel/labour allowances are in OPEX.

Using the illustrative room duty, clinic electricity costs 2,180 [D] BDT/year before fixed fees and taxes. This modest energy bill constrains an energy-only product price. Load-shedding coincidence with useful wind is NOT FOUND. Neither a mortality-reduction total nor a fish-yield uplift is fabricated.

2 Solution, engineering boundary and simulation audit

The mechanical core captures wind with a spiral rotor and transmits rotation through shafts and bevel gears to either a fan or paddles. The combined physical case in this report is a separate roof installation plus a separate pond installation; their energy is not counted twice. The advanced website adds PV, battery, motor, controls and freewheels; it is not an electricity-free extension. [35, 36, 39]

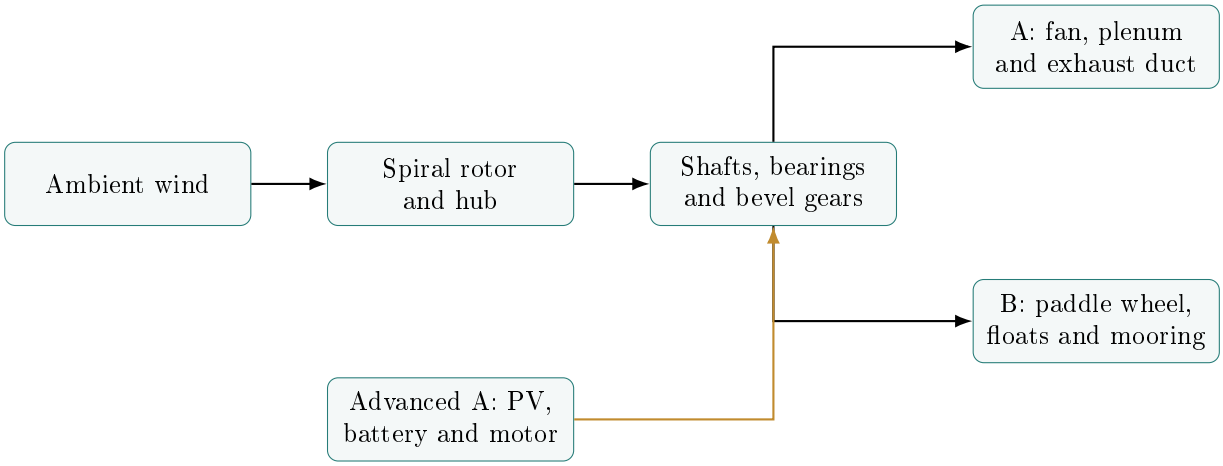


Figure 1: System pathways [S] project concept, with [A] production protection and advanced assistance boundary.

2.1 What the code computes

For the basic geometry, $P_s = \min(\frac{1}{2}\rho A v^3 C_p \eta_d, P_r)$ above the assumed starting speed; $Q = P_s \eta_f \, 3600/\Delta p$. The room model is a perfectly mixed equilibrium CO2 balance. The pond model applies standard oxygen transfer, a temperature/saturation correction, background reaeration and constant oxygen demand. It does not integrate a whole-night oxygen budget. [39]

Parameter	Basic / pond default	Advanced default	Basis
Swept rotor size	1.0 m diameter	3.46 m^2 area	[A] Different geometries
Power coefficient / drivetrain efficiency	0.18 / 0.75	0.23 / 0.84	[A] Untested coefficients
Fan model	20 Pa; 0.35 efficiency	60 W at 240 RPM	[A] Not a certified curve
Room / pond volume	150 / 100 m^3	Same scene; separate model	[A] Default sizes
CO2 / DO targets	1,200 ppm / 4 mg/L	No clinical certification	[A] Demo targets
Pond standard SAE / demand	0.80 kg/kWh; 0.25 mg/L/h	Not implemented	[A] Transfer model
PV / battery / motor	None for uninstrumented core	180 W / 240 Wh / 95 W input	[A] Site concept
Battery lock / recovery	No battery	12% / 20%	[A] Control thresholds

Parameter	Basic / pond default	Advanced default	Basis
Assistance on/off / verification	No motor	0.90 / 1.05 target; 1.5 s	[A] Hysteresis

2.2 Every weather case and operating demonstration

Mode	Case	Wind / pattern	What it does not establish	Basis
Basic A	Rooftop breeze	4.5 m/s; steady	Steady illustrative operation; no measured power curve.	[A] [39]
Basic A	Urban canyon	2.8 m/s; lull	Shelter/lulls; no urban turbulence or fatigue validation.	[A] [39]
Basic A	Heatwave	2.3 m/s; gusts	Heat exposure warnings; no refrigeration or humidity guarantee.	[A] [39]
Basic A	Monsoon	7.0 m/s; gusts	Rain effects and optional damper; not waterproof certification.	[A] [39]
Basic A	Cyclone	19.0 m/s; storm	Display continues spinning; no automatic safe cut-out.	[A] [39]
Pond B	Park breeze	4.5 m/s; steady	Steady illustrative operation; no measured power curve.	[A] [39]
Pond B	Sheltered lull	2.8 m/s; lull	Repeating lull is synthetic, not a weather time series.	[A] [39]
Pond B	Hot night	2.2 m/s; gusts	Pre-dawn oxygen risk; no survival or transient DO proof.	[A] [39]
Pond B	Monsoon	7.0 m/s; gusts	Rain effects and optional damper; not waterproof certification.	[A] [39]
Pond B	Storm surge	19.0 m/s; storm	Display continues spinning; no automatic safe cut-out.	[A] [39]

Advanced control / state	Demonstration	Limit	Basis
Sunny / sun position	PV drives or charges while the illustrated tracker follows sun position.	No measured tracking gain, shading or thermal model.	[A] [39]
Canyon	Sheltered wind with solar assistance.	No site-specific solar/wind joint distribution.	[A]
Monsoon	PV and motor assistance are disabled by policy.	Physical PV is not inherently zero in every overcast condition.	[A]
Adequate / weak / no wind	Wind-only, shared torque, or motor-driven fan through freewheels.	No clutch durability or measured torque curve.	[A]
Low battery / night	Hysteretic motor lockout; battery drains with no PV at night.	Short autonomy; no guaranteed night service.	[A]
Injected fault / wind-only	Motor inhibited, mechanical wind path remains.	No certified functional-safety system.	[A]
Pause / speed / section / explode / tour	Presentation and inspection controls; exploded flow paths hidden.	No fabrication geometry or independent engineering proof.	[A]

Advanced control / state	Demonstration	Limit	Basis
Moving / moored pond boats	Changes visual mounting and unit count.	No mooring loads, navigation, capsize or hydrodynamic validation.	[A]

Wind profile selectors include steady, gusts, repeating lull and storm. Hardware choices cover cup/ultrasonic or Hall wind sensing; MQ-135, MH-Z19B, SCD41 or industrial CO2; analog/galvanic/optical DO; rain and water-temperature probes; ATtiny, Uno, Pico, ESP32 or STM32; and open, IP54 or IP67 enclosures. The code applies temperature, ingress and power assumptions. Sensor ratings are not verified datasheets. Electronics require an external power supply, and warnings do not replace a mechanical brake. A complete literal/parameter inventory is retained in the repository audit. [39]

At the basic rooftop breeze preset, the code gives 5.92 [D] W and 372.8 [D] m^3/h, or 2.49 [D] added ACH. These are calculated examples, not measurements. The advanced rated-flow assumption yields 5.67 [D] added ACH. The illustrative battery from its default charge to lockout supplies only 1.87 [D] hours at nominal motor demand without wind, PV or parasitic loads.

Older naturally ventilated clinic guidance discussed in the clinic study implies 1,728 [D] m^3/h for the default occupancy. The financial example below targets a smaller incremental flow; it cannot be represented as compliance with that guidance. Real intake pollution, thermal comfort, filtration pressure loss and room mixing require assessment. [12, 47]

3 Components, market evidence and production costs

The baseline prototype uses PLA, but a production candidate uses supported metal shafts/gears, protected structure and inspectable floats. Proposed dimensions and quantities are design assumptions pending stress, thermal and water tests. Catalogue gear selection depends on module, ratio, material, bore and permitted loads; a cheap model gear is not an equivalent part. [36, 33, 48]

3.1 Observed prices and the three-quote test

Family / supplier	Visible price	BDT / USD	Caveat and basis
PLA filament kg / HTE Bangladesh	1,287.00 BDT	1,287.00 / 10.41	open text; out of stock on recheck; [S] indication, [D] FX [23]
PLA filament kg / Costareen	1,550.00 BDT	1,550.00 / 12.54	dated guide starting price; [S] indication, [D] FX [21]
PLA filament kg / Electronics.Com.BD	1,999.00 BDT	1,999.00 / 16.17	open text; [S] indication, [D] FX [24]
PLA filament kg / TechShopBD	1,950.04 BDT	1,950.04 / 15.77	open marketplace; [S] indication, [D] FX [22]
UCP204 bearing / Daraz/Fair Enterprise	450.00 BDT	450.00 / 3.64	cached stock conflicting; [S] indication, [D] FX [9]
UCP204 bearing / MISUMI	15.65 SGD	1,511.56 / 12.22	search-index only; open 429; [S] indication, [D] FX [10]
UCP204 bearing / Motion	52.70 USD	6,516.36 / 52.70	open text; [S] indication, [D] FX [11]
Electric exhaust fan / Walton	1,250.00 BDT	1,250.00 / 10.11	open full text; index showed 1310; [S] indication, [D] FX [17]
Hobby bevel gear / ZerOneTech	245.00 BDT	245.00 / 1.98	search-index only; open failed; [S] indication, [D] FX [27]
Conventional aerator / NSC BD	50,000.00 BDT	50,000.00 / 404.37	open text; [S] indication, [D] FX [1]
Conventional aerator / Bangla Ltd.	40,000.00 BDT	40,000.00 / 323.49	open text; inconsistent transfer claims; [S] indication, [D] FX [4]
Conventional aerator / BDStall single seller	40,000.00 BDT	40,000.00 / 323.49	open text; seller independence not established; [S] indication, [D] FX [8]
Battery 12V20Ah / CityTechBD	1,845.00 BDT	1,845.00 / 14.92	open text; suspicious outlier; [S] indication, [D] FX [5]
Battery 12V20Ah / Teexla	8,200.00 BDT	8,200.00 / 66.32	open text; [S] indication, [D] FX [6]

Prices were observed on 2026-10-22. Listings without publication dates are undated; stock and cached prices can differ. The Walton full page displayed a lower price than its indexed result; the opened full-page price is used. Foreign prices are not landed Bangladesh quotes. MISUMI and the hobby-gear price remain index-only and low-confidence. The low battery listing is an outlier and is excluded from the warranted production battery allowance.

Family	Seller count	Low / median / high BDT	Basis / limitation
Battery 12V20Ah	2	1,845.00 / 5,022.50 / 8,200.00	Three comparable independent landed quotes NOT FOUND; [D] indication summary
Conventional aerator	2	40,000.00 / 40,000.00 / 50,000.00	Three comparable independent landed quotes NOT FOUND; [D] indication summary

Family	Seller count	Low / median / high BDT	Basis / limitation
Electric exhaust fan	1	1,250.00 / 1,250.00 / 1,250.00	Three comparable independent landed quotes NOT FOUND; [D] indication summary
Hobby bevel gear	1	245.00 / 245.00 / 245.00	Three comparable independent landed quotes NOT FOUND; [D] indication summary
PLA filament kg	4	1,287.00 / 1,750.02 / 1,999.00	Family-level comparison only; brands differ; [D] indication summary
UCP204 bearing	3	450.00 / 1,511.56 / 6,516.36	Three comparable independent landed quotes NOT FOUND; [D] indication summary

Only filament has multiple independent, broadly comparable local price observations. Generic and branded bearing grades differ. Three current, independent production quotations for each bespoke component are NOT FOUND. The BOM low/base/high columns below are allowances [A], not a substituted quotation median. Supplier statements about oxygen output also differ internally; they are not accepted as field SAE. [4, 19]

3.2 Costed production BOM for each application

3.3 A: basic rooftop ventilation

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Spiral rotor and balanced hub. Formed aluminium; 1 m nominal basic / 2.1 m hybrid	1	3,500 / 6,000 / 10,000	Capture wind; [A], production quotes NOT FOUND
Shaft stock and machining. Steel, nominal 20 mm; three axes for two stages, two for H	3	1,500 / 2,700 / 4,800	Transmit torque; [A], production quotes NOT FOUND
Production bevel gear pair. Steel, two right-angle stages; module and rating RFQ	2	1,400 / 3,000 / 6,000	Redirect rotation; [A], production quotes NOT FOUND
Radial bearing blocks. 20 mm sealed UCP-type; two radial supports per shaft; axial separate	6	2,700 / 5,400 / 12,000	Support shaft; [A], production quotes NOT FOUND
Thrust bearing and collar set. Matched axial load and positive retention	1	500 / 1,000 / 2,000	Carry fan/gear thrust; [A], production quotes NOT FOUND
Couplings keys collars. Keyed steel; fastener locking	3	600 / 1,200 / 2,100	Locate and couple shafts; [A], production quotes NOT FOUND
Frame mounting and isolators. Coated local steel; engineered anchor plates	1	2,500 / 4,500 / 7,500	Structural support; [A], production quotes NOT FOUND
Fail-safe brake and storm restraint. Mechanical brake/furling concept; test required	1	1,000 / 2,500 / 5,000	Overspeed protection; [A], production quotes NOT FOUND
Rotor and gear guards. Corrosion-resistant mesh with access covers	1	1,000 / 1,800 / 3,500	Entanglement protection; [A], production quotes NOT FOUND

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Seals grease fasteners coating. Gaskets, compatible lubricant, corrosion isolation	1	700 / 1,200 / 2,200	Weather and service kit; [A], production quotes NOT FOUND
Mechanical axial fan and hub. NACA 65-810 concept; needs pressure-flow test	1	1,200 / 2,200 / 4,000	Exhaust room air; [A], production quotes NOT FOUND
Sleeve plenum duct and cowl. Sheet metal; screened intake, access and rain drain	1	1,800 / 3,500 / 6,500	Air pathway; [A], production quotes NOT FOUND
Roof flashing and backdraft damper. Waterproof penetration and service access	1	800 / 1,500 / 3,000	Weatherproof building interface; [A], production quotes NOT FOUND
Assembly balancing and inspection. Paid shop labour and acceptance test	1	2,000 / 3,500 / 6,000	Deliver inspected ex-works unit; [A], production quotes NOT FOUND

3.4 B: pond aeration

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Spiral rotor and balanced hub. Formed aluminium; 1 m nominal basic / 2.1 m hybrid	1	3,500 / 6,000 / 10,000	Capture wind; [A], production quotes NOT FOUND
Shaft stock and machining. Steel, nominal 20 mm; three axes for two stages, two for H	3	1,500 / 2,700 / 4,800	Transmit torque; [A], production quotes NOT FOUND
Production bevel gear pair. Steel, two right-angle stages; module and rating RFQ	2	1,400 / 3,000 / 6,000	Redirect rotation; [A], production quotes NOT FOUND
Radial bearing blocks. 20 mm sealed UCP-type; two radial supports per shaft; axial separate	6	2,700 / 5,400 / 12,000	Support shaft; [A], production quotes NOT FOUND
Thrust bearing and collar set. Matched axial load and positive retention	1	500 / 1,000 / 2,000	Carry fan/gear thrust; [A], production quotes NOT FOUND
Couplings keys collars. Keyed steel; fastener locking	3	600 / 1,200 / 2,100	Locate and couple shafts; [A], production quotes NOT FOUND
Frame mounting and isolators. Coated local steel; engineered anchor plates	1	2,500 / 4,500 / 7,500	Structural support; [A], production quotes NOT FOUND
Fail-safe brake and storm restraint. Mechanical brake/furling concept; test required	1	1,000 / 2,500 / 5,000	Overspeed protection; [A], production quotes NOT FOUND
Rotor and gear guards. Corrosion-resistant mesh with access covers	1	1,000 / 1,800 / 3,500	Entanglement protection; [A], production quotes NOT FOUND
Seals grease fasteners coating. Gaskets, compatible lubricant, corrosion isolation	1	700 / 1,200 / 2,200	Weather and service kit; [A], production quotes NOT FOUND
Paddle wheel and wet shaft guards. HDPE paddles on corrosion-resistant hub	1	1,800 / 3,500 / 6,000	Surface aeration; [A], production quotes NOT FOUND

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Compartmented pontoons. Recovered/rotomoulded HDPE, two separate floats	2	2,400 / 4,400 / 8,000	Flotation with reserve buoyancy; [A], production quotes NOT FOUND
Mooring anchor retrieval and wet seals. Two lines, anchors and dry access provision	1	1,200 / 2,500 / 5,000	Retain and retrieve pond assembly; [A], production quotes NOT FOUND
Assembly balancing and inspection. Paid shop labour and acceptance test	1	2,000 / 3,500 / 6,000	Deliver inspected ex-works unit; [A], production quotes NOT FOUND

3.5 Combined A+B: two separate installed assemblies

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Spiral rotor and balanced hub. Formed aluminium; 1 m nominal basic / 2.1 m hybrid	2	7,000 / 12,000 / 20,000	Capture wind; [A], production quotes NOT FOUND
Shaft stock and machining. Steel, nominal 20 mm; three axes for two stages, two for H	6	3,000 / 5,400 / 9,600	Transmit torque; [A], production quotes NOT FOUND
Production bevel gear pair. Steel, two right-angle stages; module and rating RFQ	4	2,800 / 6,000 / 12,000	Redirect rotation; [A], production quotes NOT FOUND
Radial bearing blocks. 20 mm sealed UCP-type; two radial supports per shaft; axial separate	12	5,400 / 10,800 / 24,000	Support shaft; [A], production quotes NOT FOUND
Thrust bearing and collar set. Matched axial load and positive retention	2	1,000 / 2,000 / 4,000	Carry fan/gear thrust; [A], production quotes NOT FOUND
Couplings keys collars. Keyed steel; fastener locking	6	1,200 / 2,400 / 4,200	Locate and couple shafts; [A], production quotes NOT FOUND
Frame mounting and isolators. Coated local steel; engineered anchor plates	2	5,000 / 9,000 / 15,000	Structural support; [A], production quotes NOT FOUND
Fail-safe brake and storm restraint. Mechanical brake/furling concept; test required	2	2,000 / 5,000 / 10,000	Overspeed protection; [A], production quotes NOT FOUND
Rotor and gear guards. Corrosion-resistant mesh with access covers	2	2,000 / 3,600 / 7,000	Entanglement protection; [A], production quotes NOT FOUND
Seals grease fasteners coating. Gaskets, compatible lubricant, corrosion isolation	2	1,400 / 2,400 / 4,400	Weather and service kit; [A], production quotes NOT FOUND
Mechanical axial fan and hub. NACA 65-810 concept; needs pressure-flow test	1	1,200 / 2,200 / 4,000	Exhaust room air; [A], production quotes NOT FOUND
Sleeve plenum duct and cowl. Sheet metal; screened intake, access and rain drain	1	1,800 / 3,500 / 6,500	Air pathway; [A], production quotes NOT FOUND
Roof flashing and backdraft damper. Waterproof penetration and service access	1	800 / 1,500 / 3,000	Weatherproof building interface; [A], production quotes NOT FOUND

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Paddle wheel and wet shaft guards. HDPE paddles on corrosion-resistant hub	1	1,800 / 3,500 / 6,000	Surface aeration; [A], production quotes NOT FOUND
Compartmented pontoons. Recovered/rotomoulded HDPE, two separate floats	2	2,400 / 4,400 / 8,000	Flotation with reserve buoyancy; [A], production quotes NOT FOUND
Mooring anchor retrieval and wet seals. Two lines, anchors and dry access provision	1	1,200 / 2,500 / 5,000	Retain and retrieve pond assembly; [A], production quotes NOT FOUND
Assembly balancing and inspection. Paid shop labour and acceptance test	2	4,000 / 7,000 / 12,000	Deliver inspected ex-works unit; [A], production quotes NOT FOUND

3.6 H: advanced hybrid ventilation

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Spiral rotor and balanced hub. Formed aluminium; 1 m nominal basic / 2.1 m hybrid	1	3,500 / 6,000 / 10,000	Capture wind; [A], production quotes NOT FOUND
Shaft stock and machining. Steel, nominal 20 mm; three axes for two stages, two for H	2	1,000 / 1,800 / 3,200	Transmit torque; [A], production quotes NOT FOUND
Production bevel gear pair. Steel, two right-angle stages; module and rating RFQ	1	700 / 1,500 / 3,000	Redirect rotation; [A], production quotes NOT FOUND
Radial bearing blocks. 20 mm sealed UCP-type; two radial supports per shaft; axial separate	4	1,800 / 3,600 / 8,000	Support shaft; [A], production quotes NOT FOUND
Thrust bearing and collar set. Matched axial load and positive retention	1	500 / 1,000 / 2,000	Carry fan/gear thrust; [A], production quotes NOT FOUND
Couplings keys collars. Keyed steel; fastener locking	3	600 / 1,200 / 2,100	Locate and couple shafts; [A], production quotes NOT FOUND
Frame mounting and isolators. Coated local steel; engineered anchor plates	1	2,500 / 4,500 / 7,500	Structural support; [A], production quotes NOT FOUND
Fail-safe brake and storm restraint. Mechanical brake/furling concept; test required	1	1,000 / 2,500 / 5,000	Overspeed protection; [A], production quotes NOT FOUND
Rotor and gear guards. Corrosion-resistant mesh with access covers	1	1,000 / 1,800 / 3,500	Entanglement protection; [A], production quotes NOT FOUND
Seals grease fasteners coating. Gaskets, compatible lubricant, corrosion isolation	1	700 / 1,200 / 2,200	Weather and service kit; [A], production quotes NOT FOUND
Mechanical axial fan and hub. NACA 65-810 concept; needs pressure-flow test	1	1,200 / 2,200 / 4,000	Exhaust room air; [A], production quotes NOT FOUND
Sleeve plenum duct and cowl. Sheet metal; screened intake, access and rain drain	1	1,800 / 3,500 / 6,500	Air pathway; [A], production quotes NOT FOUND

Component, material/specification	Qty	Low / base / high BDT	Function / basis
Roof flashing and backdraft damper. Waterproof penetration and service access	1	800 / 1,500 / 3,000	Weatherproof building interface; [A], production quotes NOT FOUND
PV module and fixed rack. 180-200 W class; roof area/structure assessment	1	6,000 / 9,000 / 15,000	Electrical assistance energy; [A], production quotes NOT FOUND
Battery pack and BMS. 24 V nominal, 240 Wh class; warranted matched pack	1	8,000 / 14,000 / 22,000	Short duration assistance; [A], production quotes NOT FOUND
Motor PWM belt pulleys dual freewheels. 95 W input class, compatible ratios	1	5,000 / 9,000 / 15,000	Assist common shaft; [A], production quotes NOT FOUND
Controller fuses disconnect cabling. Weatherproof DC controls and two tachometers	1	3,500 / 6,000 / 10,000	Protected DC bus; [A], production quotes NOT FOUND
Sun tracker structure and actuators. Optional site depiction; costed only in H	1	2,500 / 5,000 / 10,000	Illustrative solar tracking; [A], production quotes NOT FOUND
Assembly balancing and inspection. Paid shop labour and acceptance test	1	2,000 / 3,500 / 6,000	Deliver inspected ex-works unit; [A], production quotes NOT FOUND

Case	Low BDT	Base BDT	High BDT	Base USD / basis
A	21,200	40,000	74,600	323.49; [D] from [A] BOM
B	22,800	43,200	80,100	349.37; [D] from [A] BOM
H	44,100	78,800	138,000	637.28; [D] from [A] BOM
A+B	44,000	83,200	154,700	672.87; [D] from [A] BOM

Ex-works totals include assembly/balancing/inspection. Sold-unit on-site installation is a separate fee and direct cost; pond access and difficult roofs may exceed that allowance. Optional monitoring is excluded from the uninstrumented core and costed in future scope. Two-site A+B duplicates rotors, shafts, gears, bearings, brakes and frames. No electrical generation revenue is assumed.

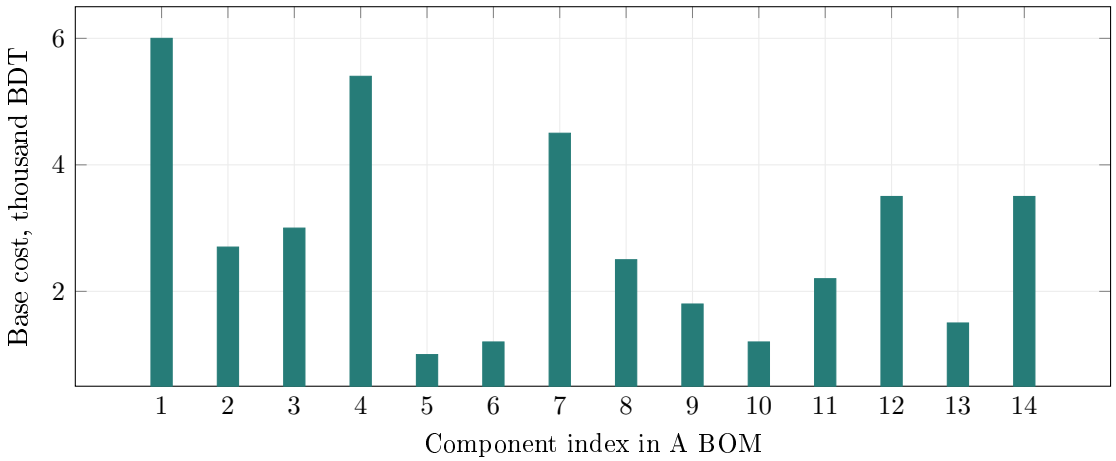


Figure 2: Base A cost breakdown [D] from [A] component allowances. Indices follow the BOM order; custom rotor, structure and safety costs matter alongside gears.

4 Sustainable fabrication and build-versus-assemble

A low-cost process should preserve predictable failure behaviour and repair access. Print patterns and noncritical covers; fabricate balanced rotor sections and structural steel locally; purchase matched bearings and rated drive parts. Recycled aluminium needs traceable alloy and fatigue checks. Recovered HDPE requires contamination and leak inspection. Bamboo is a replaceable noncritical jig/support candidate rather than an assumed structural roof anchor. Material choices require field tests, and neither PLA nor PETG is assumed to harmlessly degrade in a pond. [34, 2, 38, 33]

Method / process	Low / base / high BDT	Tooling / boundary	Basis
Printed pattern + formed recycled aluminium rotor. Print segmented pattern, roll/form sheet, rivet, balance	3,000 / 6,000 / 11,000	Shared jigs; alloy and fatigue evidence required	[A] Alternative process budgets; overlap BOM
Purchased gears / local steel shaft and frame. Machine/key shaft, weld and coat frame, align purchased bearings	5,000 / 10,000 / 19,000	Lathe, welding, alignment jig; paid labour and scrap	[A] Alternative process budgets; overlap BOM
Recycled PLA / PETG noncritical parts per unit. Dry, filter, print coupons then ducts/covers only	800 / 1,800 / 3,500	Reuse existing printer; no critical strength inferred	[A] Alternative process budgets; overlap BOM
Reclaimed HDPE float pair. Inspect contamination, leak-test compartments, mechanically cradle	2,400 / 4,400 / 8,000	No toxic drums, no glued untested seams	[A] Alternative process budgets; overlap BOM
Bamboo replaceable noncritical fixtures. Dry, treat appropriately, bolt and inspect for rot	500 / 1,500 / 3,000	No untested roof anchor substitution	[A] Alternative process budgets; overlap BOM
Recovered metal and polymer take-back per unit. Label material, separate fasteners, recover at end of life	300 / 700 / 1,500	Recovery target, recycler agreement NOT FOUND	[A] Alternative process budgets; overlap BOM

Recycled-polymer research found altered properties and greater variability after reprocessing. It supports coupon testing rather than a blanket claim of equal lifetime. Supplier material guidance supports screening for outdoor UV/temperature suitability; it is not structural qualification for this application. These sources are foreign, and the recycling paper exceeds the age threshold. [38, 34, 2]

4.1 Batch economics

Per-unit cost is $C(q) = V + F/q$ and the crossover is $q^* = \lceil (F_{make} - F_{assemble})/(V_{assemble} - V_{make}) \rceil$. Here "make means locally fabricate the custom rotor/frame while still buying precision drive elements. Making bearing balls or cutting all gears in-house is not assumed economical. Setup, variable savings, scale capability and rejects are assumptions; crossover is not a sales forecast.

Mode	Batch units	Make BDT/unit	Assemble BDT/unit	Crossover / basis
A	1	212,800	70,000	21 units; [D] from [A] setup/rates
A	10	50,800	43,000	21 units; [D] from [A] setup/rates
A	100	34,600	40,300	21 units; [D] from [A] setup/rates
A	1,000	32,980	40,030	21 units; [D] from [A] setup/rates
B	1	215,424	73,200	20 units; [D] from [A] setup/rates
B	10	53,424	46,200	20 units; [D] from [A] setup/rates
B	100	37,224	43,500	20 units; [D] from [A] setup/rates
B	1,000	35,604	43,230	20 units; [D] from [A] setup/rates
A+B	1	428,224	143,200	21 units; [D] from [A] setup/rates
A+B	10	104,224	89,200	21 units; [D] from [A] setup/rates
A+B	100	71,824	83,800	21 units; [D] from [A] setup/rates
A+B	1,000	68,584	83,260	21 units; [D] from [A] setup/rates
H	1	364,616	128,800	18 units; [D] from [A] setup/rates
H	10	94,616	83,800	18 units; [D] from [A] setup/rates
H	100	67,616	79,300	18 units; [D] from [A] setup/rates
H	1,000	64,916	78,850	18 units; [D] from [A] setup/rates

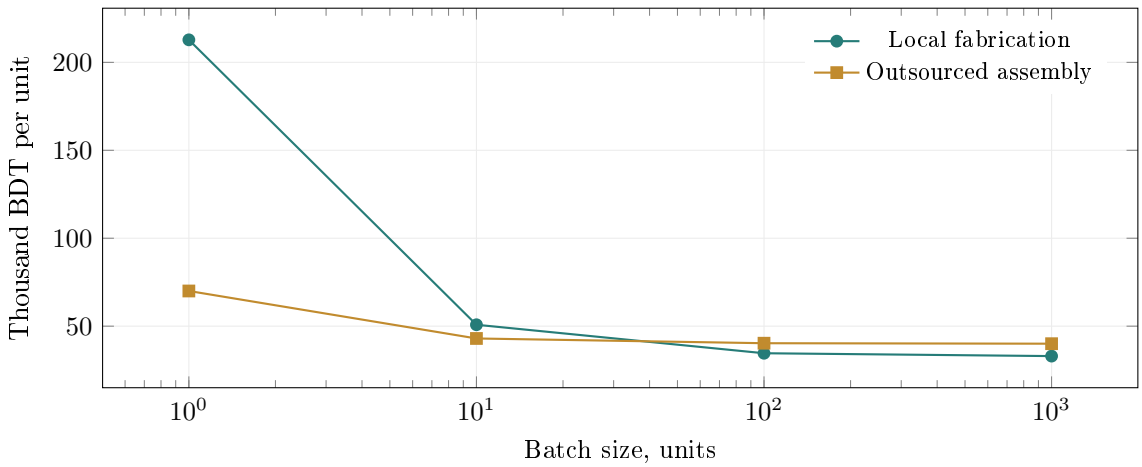


Figure 3: Build versus assemble for A [D] from [A] setup/variable allowances. Precision parts remain purchased in both options.

5 Equivalent-service comparison and customer affordability

The cost test holds the useful service boundary explicit. Basic ventilation supplies 300 [A] m^3/h of added flow to a 150 [A] m^3 room for 4,380 [A] h/year; the conventional fan draw is 55 [S] W, but the required pressure-flow duty is unmeasured. The pond example needs 0.025 [D] kg O2/h continuously. Electric field SAE 1.2 [A] kg/kWh and wind field correction 0.5 [A] remain assumptions. [17, 19, 39]

FAO historical foreign tests show example paddlewheel specific oxygen transfer from 0.91 [S] to 1.41 [S] kg/kWh under their stated conditions; modern supplier nameplates differ. Neither is a clean substitute for a measured field oxygen-transfer curve. The commercial aerator used as a capex reference is larger than the small example demand; energy is duty-normalised, with mixing and start/stop feasibility unresolved. A smaller conventional unit could be cheaper, further weakening the wind case. [19, 1, 4]

Wind-only systems have an unmet-service fraction. Equal-service costs therefore retain the conventional machine for backup and charge wind equipment maintenance and replacements on top. The cost of the existing grid connection is common to both alternatives. These totals assume available backup power; outage resilience still needs a generator/battery dispatch and simultaneous-weather assessment. Wind/PV substitution is not a guaranteed hourly service.

Base case	Electric, BDT	Wind + backup, BDT	Diesel-fed, BDT	Basis
A	50,002	106,569	198,996	[D] undiscounted horizon totals from [A/S]
H	50,002	211,770	198,996	[D] undiscounted horizon totals from [A/S]
B	153,032	236,983	285,300	[D] undiscounted horizon totals from [A/S]

The verified commercial flat tariff is 15.36 [S] BDT/kWh and clinic LT-D1 is 9.05 [S] BDT/kWh under the June revision. The scanned official order and correction were inspected visually. Future tariffs are not forecast. VAT and fixed demand charges are not credited as avoidable energy savings on an existing connection. [44, 45]

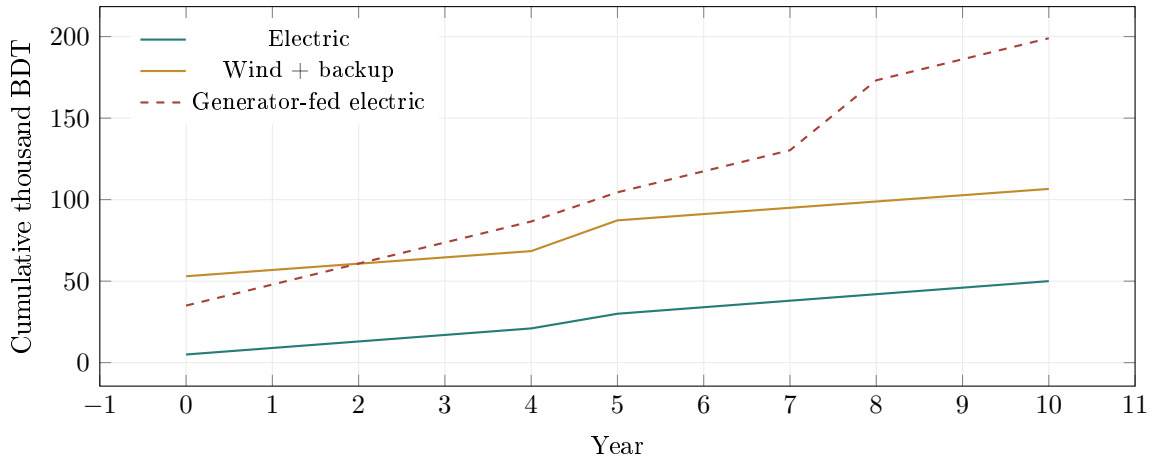


Figure 4: Equal-service cumulative ventilation cost [D] using [A/S] inputs. Generator sharing/fuel assumptions are weak; wind operation alone has unmet demand.

5.1 Basic A versus advanced H, and ventilation versus aeration

Criterion	Basic A	Advanced H	Pond B / basis
Principal service	Room air exchange	Wind/PV/battery room air exchange	Supplemental oxygen/mixing; [S/A] concepts
Failure during calm	Fan backup required	Limited battery; backup still required	Dangerous pre-dawn DO gap; [A] risk
Environmental controls	Intake pollution, rain and pressure	Electrical protection and battery replacement	Corrosion, leaks, mooring, aquatic fragments; [A]
Market entry	Supervised noncritical test room	More capital and service skills	Noncritical ornamental pond before stocked aquaculture; [A] judgement
Health/yield value	Not monetised	Not monetised	Not monetised; measurements NOT FOUND
Wind dependency	Rooftop turbulence	Solar can assist subject to weather/storage	Pond-height shelter and correlated night lulls; [A]

Case	Wind energy fraction	Annual net energy saving, BDT	Incremental payback	Basis
A	47.1%	143.4	334.8	[D] after wind maintenance, before health/yield value
H	85.0%	−6.8	No finite / beyond decision horizon	[D] after wind maintenance, before health/yield value
B	7.6%	−1,515.1	No finite / beyond decision horizon	[D] after wind maintenance, before health/yield value

Case / denominator	Electric cost	Wind + backup cost	Basis
A / BDT per added ACH-hour	0.571	1.217	[D] total cost / total service, same room volume or oxygen demand
H / BDT per added ACH-hour	0.571	2.417	[D] total cost / total service, same room volume or oxygen demand
B / BDT per field kg O2	69.878	108.212	[D] total cost / total service, same room volume or oxygen demand

This is an unsubsidised customer comparison. The additional-flow target is not a full clinic safety specification. Producer profit can coexist with poor customer economics if tariffs are set above avoided energy cost; the report does not call that affordability. Subsidies or sponsor purchases may fund demonstration value, but are not evidence of a sustainable mass-market saving.

6 Wind resource, performance availability and market boundary

The national NREL campaign establishes a credible measurement/model framework, not a power curve for this prototype. The EU study indexed appendix reports Chandpur at 4.11 [S] m/s at 60 [S] m, based on the older campaign. Full access to that PDF was blocked, so this contextual number is low-confidence and not used as a rooftop input. Roof turbulence and pond-height shelter need local measurement. [49, 16, 48]

The screening uses a Weibull wind distribution with shape $k = 2$ [A], $c = \overline{v}/\Gamma(1 + 1/k)$ [D], an assumed cut-in and a real-operation cut-out absent from the display model. Numerical integration evaluates both the probability of meeting required shaft power and $E[\min(P_s/P_{needed}, 1)]$. Mechanical availability multiplies both. A capacity factor relative to the displayed rating is a separate quantity. These are illustrative scenario integrals, not bankable wind-resource forecasts.

Scenario	Mean wind m/s	Full service time	Energy fraction	Rated CF	Basis
conservative	2.5	8.8%	24.1%	0.6%	[D] from [A] distribution, coefficients and uptime
base	3.5	29.2%	47.1%	1.9%	[D] from [A] distribution, coefficients and uptime
optimistic	4.5	48.1%	63.6%	4.3%	[D] from [A] distribution, coefficients and uptime

The required wind threshold for the airflow example is 4.19 [D] m/s under the assumed geometry and losses. The fraction of time real hardware delivers this output remains NOT FOUND. Published high-tower data cannot supply a measured rooftop/pond availability percentage. The simple annual hybrid energy allowance is separate from the website battery simulation and does not prove hourly adequacy.

Serviceable-market search funnel	Low / base / high sites	Basis
Qualified roof sites in first service territory	100 / 500 / 1,500	[A] No local qualification or willingness-to-pay survey
Qualified supplemental pond sites in first territory	50 / 250 / 1,000	[A] No local qualification or willingness-to-pay survey

National fisheries area and pond count are broad context. A qualified site must pass wind, access, safety, maintenance and payment checks. First customers should be a university demonstration facility, a sponsor-funded community test room or a municipal ornamental-pond pilot. No clinical provider, farmer or sponsor has supplied a purchase commitment in this task. Adoption is a scenario, not a percentage inferred from national hectares.

7 Investment, OPEX, revenue and profit

7.1 Prototype-to-market investment

Investment item	Low BDT	Base BDT	High BDT	Treatment / basis
Engineering and redesign	250,000	500,000	1,000,000	prelaunch; [A] lines, [D] total
Tools printer jigs metrology	200,000	450,000	900,000	capex; [A] lines, [D] total
Pilot fabrication and failed trials	150,000	300,000	600,000	prelaunch; [A] lines, [D] total
Independent performance and structural tests	150,000	350,000	800,000	prelaunch; [A] lines, [D] total
Registration legal and launch	50,000	100,000	200,000	prelaunch; [A] lines, [D] total
low total incl contingency and working capital	1,060,000	–	–	D total; [A] lines, [D] total
base total incl contingency and working capital	–	2,240,000	–	D total; [A] lines, [D] total
high total incl contingency and working capital	–	–	4,600,000	D total; [A] lines, [D] total

Totals include an explicitly budgeted contingency and working capital. Prelaunch engineering, pilots, tests, legal/launch and contingency are cash outflows at inception; tools are capitalised, working capital is an asset released at the horizon. Post-launch fleet growth and later tool renewal can raise the financing need far above the initial budget. No government grant is included in project FCF.

7.2 Fixed OPEX, monthly and annual

Item	Low / base / high monthly BDT	Base annual BDT	Basis
Engineering and founder salaries	40,000 / 60,000 / 100,000	720,000	[A] paid resources; no assumption of permanently free labour
Technician salary	18,000 / 25,000 / 40,000	300,000	[A] paid resources; no assumption of permanently free labour
Rent	8,000 / 12,000 / 20,000	144,000	[A] paid resources; no assumption of permanently free labour
Utilities and workshop energy	2,000 / 3,500 / 6,000	42,000	[A] paid resources; no assumption of permanently free labour
Logistics and field travel	4,000 / 8,000 / 15,000	96,000	[A] paid resources; no assumption of permanently free labour

Item	Low / base / high monthly BDT	Base annual BDT	Basis
Marketing and customer development	3,000 / 7,000 / 14,000	84,000	[A] paid resources; no assumption of permanently free labour
Administration accounting telecom	3,000 / 6,000 / 10,000	72,000	[A] paid resources; no assumption of permanently free labour
Insurance and liability allowance	1,000 / 3,000 / 8,000	36,000	[A] paid resources; no assumption of permanently free labour
Tool maintenance and calibration	2,000 / 4,000 / 8,000	48,000	[A] paid resources; no assumption of permanently free labour

Manufacturing direct cost, sold-unit installation, warranty reserve, provider-fleet service and maintenance-contract delivery are variable costs outside this fixed pool. The base fixed pool escalates with costs and adds a capacity allowance as the calendar ramp grows. Extra headcount required for very high break-even volumes is not validated. Insurance, roof liability and audited tax treatment require quotations/advice. Tax is an explicit profit reserve, with no refund for losses.

7.3 Proposed commercial models and tariffs

No sale, lease, pay-per-use, installation or maintenance business model was found in the supplied PDFs or current source. The following are analyst-proposed models requested for appraisal. Competitor prices anchor the opportunity cost but do not substantiate ZeroWatt willingness to pay. [36, 35, 39, 17, 1]

Model	Base tariff / transaction	Customer and billing boundary	Basis
Hardware sale	A 45,000; B 55,000; H 125,000 BDT	Site owner buys unit; installation unbundled; warranty cost retained	[A] unvalidated WTP
Lease	1,500 BDT/unit/month	Provider owns fleet; supplemental access, no guaranteed critical service	[A] includes fleet service
Pay-per-use	30 BDT/delivered hour	Scheduled 120 h/month times wind energy-service fraction; metering/collection untested	[A] provider owns fleet
Installation	3,000 BDT/sold unit	Separate charge only on sold units; no duplicate leased-asset billing	[A] direct labour allowance
Maintenance contract	1,800 BDT/year	Eligible sold units after warranty; assumed uptake 40%	[A] not also billed to lease/use fleet

Combined deployments allocate 70% [A] to sale, 20% [A] to lease and 10% [A] to metered use. Product mix is 60% [A] A, 30% [A] B and 10% [A] H. These are mutually exclusive asset channels.

Installation and post-warranty service attach only under the stated rules. Assets are not sold and leased simultaneously.

Year	New base deployments	Conservative	Optimistic	Basis
1	30	15	54	[A] ramp; [D] scenario scaling
2	45	22	81	[A] ramp; [D] scenario scaling
3	65	32	117	[A] ramp; [D] scenario scaling
4	90	45	162	[A] ramp; [D] scenario scaling
5	120	60	216	[A] ramp; [D] scenario scaling
6	150	75	270	[A] ramp; [D] scenario scaling
7	180	90	324	[A] ramp; [D] scenario scaling
8	210	105	378	[A] ramp; [D] scenario scaling
9	240	120	432	[A] ramp; [D] scenario scaling
10	270	135	486	[A] ramp; [D] scenario scaling

Expected deployments can be fractional in monthly/scenario outputs; actual orders must be integers. New units are spread uniformly across months and are not front-loaded to improve NPV. Cohort churn, collection loss and useful fleet life limit recurring revenue. There is no terminal enterprise multiple or unproven licence revenue.

7.4 Monthly and annual consolidated results

Year	Revenue BDT	EBITDA BDT	Depreciation	Tax reserve	Net profit BDT	Basis
0	0	−1,590,000	0	0	−1,590,000	[D]
1	1,288,441	−1,277,878	135,474	0	−1,413,352	[D]
2	2,174,375	−1,050,380	244,891	0	−1,295,272	[D]
3	3,439,540	−690,246	411,484	0	−1,101,730	[D]
4	5,145,112	−372,399	655,112	0	−1,027,511	[D]
5	7,361,558	132,656	997,752	0	−865,096	[D]
6	9,827,825	747,380	1,428,463	0	−681,082	[D]
7	12,532,690	1,455,905	1,898,352	0	−442,447	[D]
8	15,509,036	2,279,737	2,442,766	0	−163,029	[D]
9	18,745,335	3,202,539	3,051,453	37,772	113,315	[D]
10	22,228,920	4,206,666	3,712,604	123,515	370,546	[D]

Year	Average monthly revenue	Average monthly EBITDA	Average monthly net profit	Average monthly FCF	Basis
1	107,370	−106,490	−117,779	−141,470	[D] annual / months
2	181,198	−87,532	−107,939	−148,291	[D] annual / months
3	286,628	−57,521	−91,811	−152,644	[D] annual / months
4	428,759	−31,033	−85,626	−166,775	[D] annual / months
5	613,463	11,055	−72,091	−175,590	[D] annual / months

Year	Average monthly revenue	Average monthly EBITDA	Average monthly net profit	Average monthly FCF	Basis
6	818,985	62,282	−56,757	−220,982	[D] annual / months
7	1,044,391	121,325	−36,871	−171,801	[D] annual / months
8	1,292,420	189,978	−13,586	−162,409	[D] annual / months
9	1,562,111	266,878	9,443	−151,906	[D] annual / months
10	1,852,410	350,555	30,879	84,070	[D] annual / months

Average monthly rows are annual results divided by operating months; they do not claim every month has equal revenue. The repository includes the full inception-to-horizon monthly schedule for all scenarios, so fleet, working-capital and renewal timing can be inspected.

7.5 Revenue and profit by model

Year / business model	Annual revenue BDT	Average monthly BDT	Basis
1 / sale	1,140,720	95,060	[D] from [A] tariffs, cohort and collection inputs
1 / lease	55,329	4,611	[D] from [A] tariffs, cohort and collection inputs
1 / pay_per_use	31,282	2,607	[D] from [A] tariffs, cohort and collection inputs
1 / installation	61,110	5,092	[D] from [A] tariffs, cohort and collection inputs
1 / maintenance	0	0	[D] from [A] tariffs, cohort and collection inputs
5 / sale	5,337,924	444,827	[D] from [A] tariffs, cohort and collection inputs
5 / lease	1,053,308	87,776	[D] from [A] tariffs, cohort and collection inputs
5 / pay_per_use	595,521	49,627	[D] from [A] tariffs, cohort and collection inputs
5 / installation	285,960	23,830	[D] from [A] tariffs, cohort and collection inputs
5 / maintenance	88,844	7,404	[D] from [A] tariffs, cohort and collection inputs
10 / sale	14,612,402	1,217,700	[D] from [A] tariffs, cohort and collection inputs
10 / lease	4,107,889	342,324	[D] from [A] tariffs, cohort and collection inputs
10 / pay_per_use	2,322,526	193,544	[D] from [A] tariffs, cohort and collection inputs
10 / installation	782,807	65,234	[D] from [A] tariffs, cohort and collection inputs
10 / maintenance	403,296	33,608	[D] from [A] tariffs, cohort and collection inputs

Year / model	EBITDA annual	Net annual	EBITDA / net monthly	Basis
1 / sale	−1,200,350	−1,280,031	−100,029 / −106,669	[D] allocation below
1 / lease	−17,993	−52,174	−1,499 / −4,348	[D] allocation below
1 / pay_per_use	−9,708	−27,052	−809 / −2,254	[D] allocation below
1 / installation	−49,826	−54,095	−4,152 / −4,508	[D] allocation below
1 / maintenance	0	0	0 / 0	[D] allocation below
5 / sale	−850,523	−915,783	−70,877 / −76,315	[D] allocation below
5 / lease	598,000	−20,046	49,833 / −1,670	[D] allocation below
5 / pay_per_use	346,941	37,076	28,912 / 3,090	[D] allocation below
5 / installation	22,187	18,690	1,849 / 1,558	[D] allocation below
5 / maintenance	16,052	14,966	1,338 / 1,247	[D] allocation below
10 / sale	−542,835	−696,009	−45,236 / −58,001	[D] allocation below
10 / lease	2,834,123	388,992	236,177 / 32,416	[D] allocation below
10 / pay_per_use	1,636,849	411,468	136,404 / 34,289	[D] allocation below
10 / installation	156,384	148,178	13,032 / 12,348	[D] allocation below
10 / maintenance	122,145	117,917	10,179 / 9,826	[D] allocation below

Direct costs follow each actual channel. Shared overhead, tools and consolidated tax are allocated by revenue; provider-fleet depreciation follows lease/use deployment shares. These rows reconcile to the combined company but do not prove independent viability of installation or AMC. Separate sale-plus-installation/service, all-lease and all-use alternatives each carry the full overhead and launch budget; they must not be summed.

Standalone alternative / year	Annual revenue	Annual EBITDA	Annual net profit	Basis
sale / 1	1,716,900	−1,273,188	−1,363,188	[D] full common overhead per alternative
sale / 5	8,161,041	−917,478	−1,007,478	[D] full common overhead per alternative
sale / 10	22,569,294	−315,745	−425,244	[D] full common overhead per alternative
lease / 1	276,645	−1,300,879	−1,542,459	[D] full common overhead per alternative

Standalone alternative / year	Annual revenue	Annual EBITDA	Annual net profit	Basis
lease / 5	5,266,540	2,353,412	−762,429	[D] full common overhead per alternative
lease / 10	20,539,447	13,863,687	1,307,878	[D] full common overhead per alternative
use / 1	312,820	−1,264,704	−1,506,284	[D] full common overhead per alternative
use / 5	5,955,211	3,042,083	−87,669	[D] full common overhead per alternative
use / 10	23,225,256	16,549,497	3,322,235	[D] full common overhead per alternative

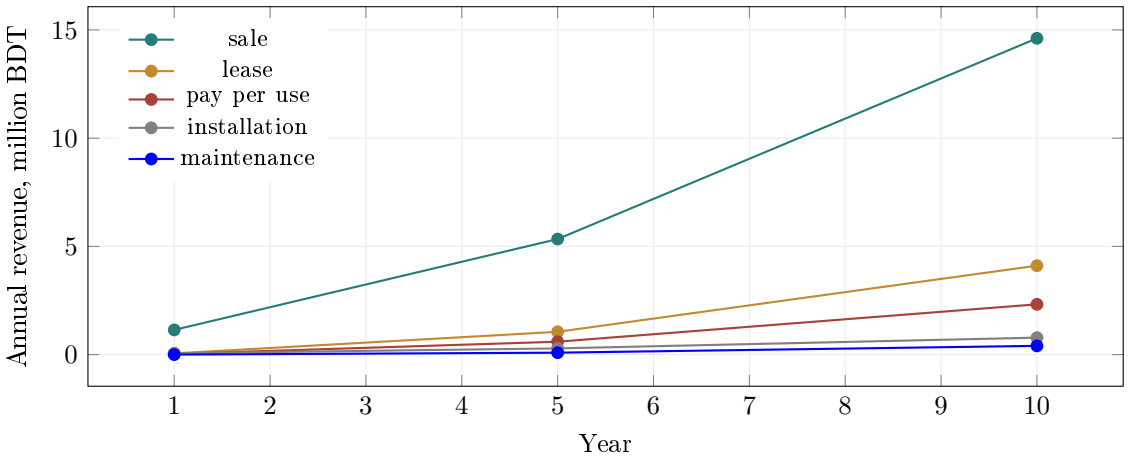


Figure 5: Base revenue by proposed model [D] from [A] tariffs and demand. Sales, leases and use have distinct asset cohorts; no double billing.

8 Cash flow, NPV, IRR and payback

The model computes $FCF = NetProfit + Depreciation - FleetCapex - ToolCapex - \Delta NWC$ and $NPV = \sum_{m=0}^{M} FCF_m/(1+r)^{m/12}$. The IRR is an annual effective rate solving the same monthly timing equation. Annual summaries are sums of monthly cash flow, not a claim of identical year-end timing. A negative IRR is meaningful when a unique root exists; no root is reported as undefined. Payback is a durable interpolated crossing that subsequent capital outlays do not reverse.

Scenario	NPV BDT / USD	IRR	Simple / discounted payback years	Peak funding BDT	Basis
conservative	−21,129,346 / −170,880	Undefined; no root	Not reached / Not reached	49,339,229	[D], [A/S] inputs
base	−11,181,078 / −90,425	−47.7%	Not reached / Not reached	21,769,906	[D], [A/S] inputs
optimistic	605,233 / 4,895	19.6%	7.85 / 9.96	6,799,321	[D], [A/S] inputs

Base year	FCF BDT	Fleet capex	Tool capex	Change in working capital	Basis
0	−2,240,000	0	450,000	200,000	[D] monthly sum
1	−1,697,638	419,760	0	0	[D] monthly sum
2	−1,779,494	654,826	0	74,288	[D] monthly sum
3	−1,831,730	983,694	0	157,790	[D] monthly sum
4	−2,001,297	1,416,519	0	212,379	[D] monthly sum
5	−2,107,080	1,964,239	0	275,497	[D] monthly sum
6	−2,651,787	2,553,511	547,494	298,163	[D] monthly sum
7	−2,061,613	3,186,782	0	330,736	[D] monthly sum
8	−1,948,906	3,866,629	0	362,014	[D] monthly sum
9	−1,822,866	4,595,764	0	391,870	[D] monthly sum
10	1,008,843	5,377,044	0	−2,302,737	[D] monthly sum

Scenario	Nominal discount rate	NPV BDT	Basis
conservative	12%	−26,789,579	[D] monthly discounting
conservative	18%	−21,129,346	[D] monthly discounting
conservative	25%	−16,757,317	[D] monthly discounting
base	12%	−13,020,405	[D] monthly discounting
base	18%	−11,181,078	[D] monthly discounting
base	25%	−9,622,555	[D] monthly discounting

Scenario	Nominal discount rate	NPV BDT	Basis
optimistic	12%	3,733,465	[D] monthly discounting
optimistic	18%	605,233	[D] monthly discounting
optimistic	25%	−1,463,703	[D] monthly discounting

The base and conservative cases fail the required return. The optimistic case assumes more adoption, higher price, cheaper components and stronger wind simultaneously; no probabilities are attached. The initial financing mix below funds only a validation budget and cannot fund the full base-case burn without repeated additional capital. Continuing the base plan would be a financing decision, not evidence that it is feasible.

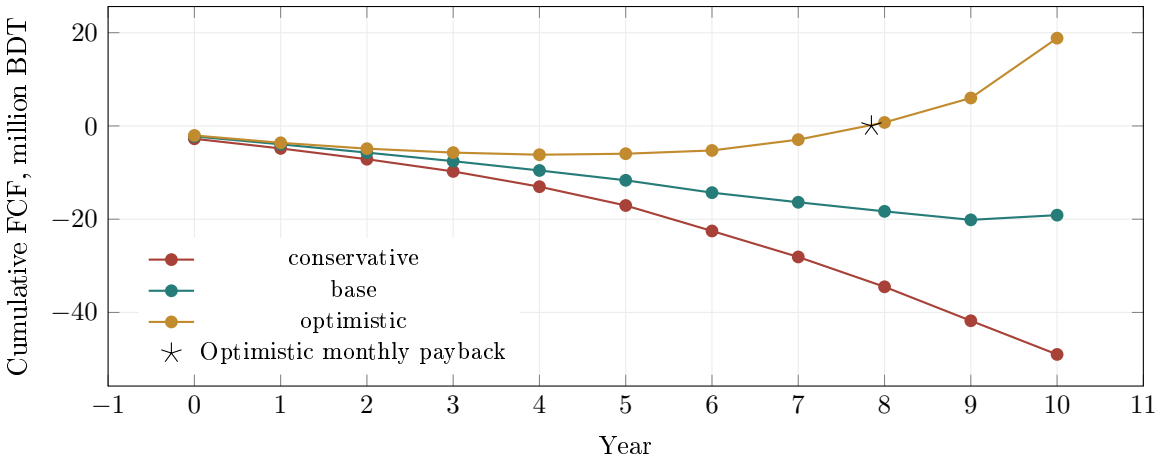


Figure 6: Cumulative company cash flow [D]. The star marks optimistic monthly interpolated payback; the base and conservative cases do not recover within the horizon. Lines join annual snapshots.

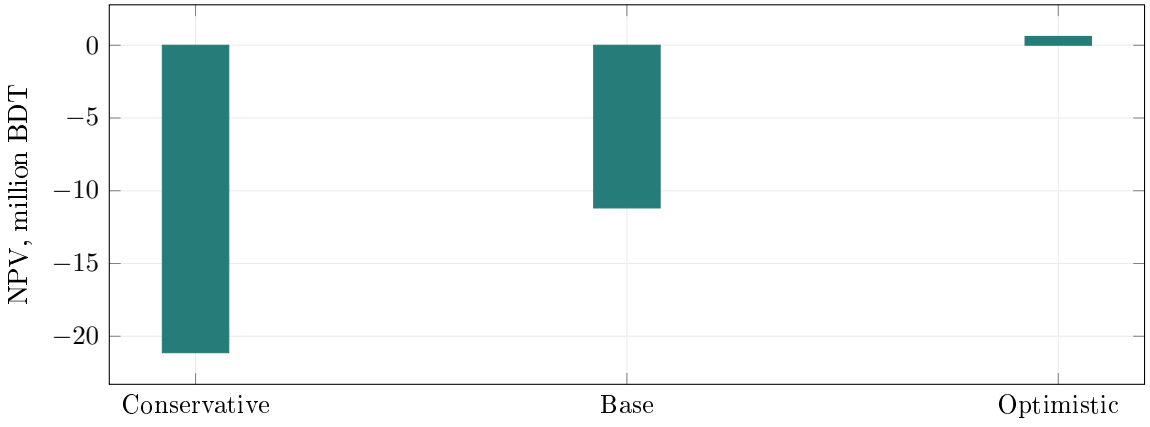


Figure 7: Scenario NPV [D] at the assumed base discount rate; positive optimistic results are conditional, not a tuned base forecast.

9 Sensitivity and break-even conditions

Input	Low / high setting	NPV at low	NPV at high	Basis
Price	{'price': 0.8} / {'price': 1.2}	−17,524,684	−5,821,504	[D] one at a time; [A] ranges
Adoption	{'adoption': 0.6} / {'adoption': 1.4}	−11,833,030	−10,832,513	[D] one at a time; [A] ranges
Wind availability	{'wind_factor': 0.6} / {'wind_factor': 1.4}	−12,263,177	−10,194,117	[D] one at a time; [A] ranges
Component cost	{'cost': 0.8} / {'cost': 1.2}	−5,679,524	−17,369,983	[D] one at a time; [A] ranges
Discount rate	.12 / .25	−13,020,405	−9,622,555	[D] one at a time; [A] ranges

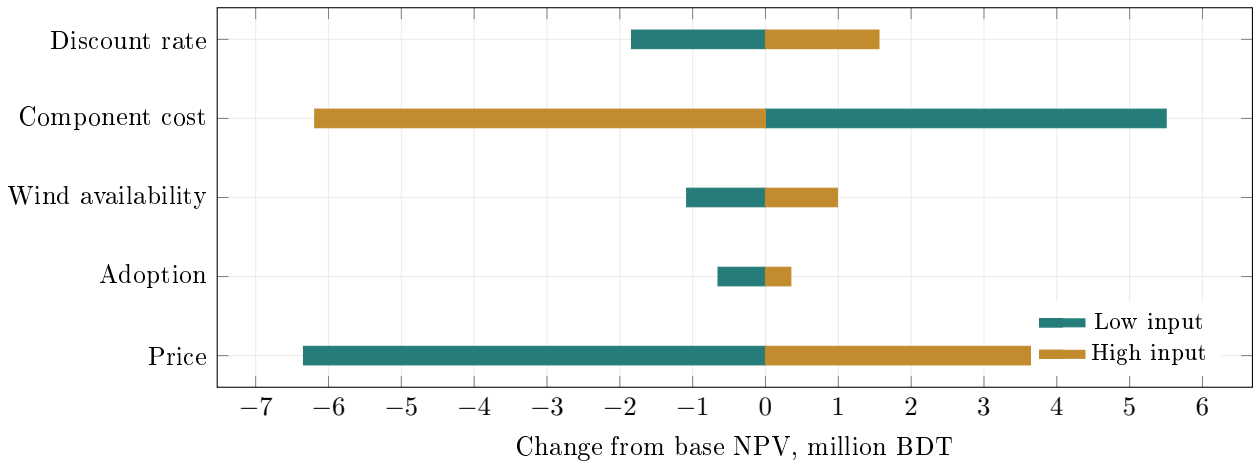


Figure 8: Tornado sensitivity [D]. Low/high describe the input, not necessarily the financial outcome. Wind variation affects metered-use hours; sale and supplemental-access lease prices are fixed.

Within the model, uniformly scaling all tariffs by 1.435 [D] makes project NPV zero; the corresponding hardware weighted price is 80,360 [D] BDT. This is a portfolio tariff condition, not a hardware-only price recommendation. Holding tariff and assumed cost capacity fixed, adoption would need 774.17 [D] times the base ramp, including 23,225.1 [D] expected first-year deployments. That scale is outside the validated staffing/customer envelope and is not a credible sales forecast.

Wind does not directly create sales or guarantee lease uptime in the proposed contracts. Accordingly the financial wind tornado is modest because it changes metered-use revenue only. A severe low-wind reality could also impair adoption, retention, warranty and liability; the one-at-a-time chart does not quantify those interactions. Capacity factor, full-service probability and billable equivalent hours are not used interchangeably.

10 Funding strategy and stage gates

Pursue non-dilutive pilot support and university facilities first, while keeping project viability separate from fundraising. Government programme amounts are ceilings/indications, not awarded money. No message or application has been sent to any organisation. Probabilities below are subjective low/base/high planning assumptions; published approval probabilities were NOT FOUND.

Route / eligibility	Amount low / base / high BDT	Cost and timing months	Success low / base / high	Basis / limitation
University / incubator. University affiliation and supervised test plan	100,000 / 400,000 / 700,000	Grant or in-kind, no assumed equity; 1 / 3 / 6	0% / 50% / 80%	A; No specific award/call verified [39]
CSR / sponsor. Instrumented community benefit and safeguarding agreement	200,000 / 700,000 / 1,500,000	Restricted pilot grant; reporting and branding; 2 / 4 / 9	0% / 25% / 50%	A; Target, not an offer [46]
iDEA / BCC / ICT Division. Confirm innovative hardware eligibility and current application session	200,000 / 700,000 / 1,000,000	Non-dilutive grant if selected; 3 / 6 / 12	0% / 15% / 30%	A/S; S ceiling 1000000, A request/time/chance [32]
Competition. Published call, judged prototype and IP terms	50,000 / 200,000 / 500,000	Prize; event-specific rules; 1 / 4 / 12	0% / 10% / 25%	A; Current prize amount/deadline NOT FOUND [32]
Angel / impact VC. Paid pilots, unit margin and defensible product	1,000,000 / 3,000,000 / 7,000,000	A equity dilution 10-30 percent; no valuation evidence; 3 / 6 / 12	0% / 5% / 15%	A; No investor commitment or term sheet [43]
Startup Bangladesh. Registered company, tax/VAT records and diligence	2,500,000 / 5,000,000 / 50,000,000	Indexed FAQ: equity/quasi-equity, at most 49 percent; confirm; 1 / 4 / 9	0% / 3% / 10%	A/S-index; FAQ indexed terms; full open blocked [43]
SME Foundation. Productive enterprise and participating lender underwriting	50,000 / 1,000,000 / 1,500,000	Published up to 9 percent; current partner terms required; 2 / 5 / 9	0% / 10% / 30%	A/S; S indicative amount/interest; A approval time/chance [40]
Bank / BB green refinance. Trade licence, accounts, eligible use and repayment capacity	500,000 / 1,500,000 / 5,000,000	A borrower rate 8-16 percent until current offer; 3 / 6 / 12	0% / 5% / 20%	A; Programme exists; product eligibility and rate NOT FOUND [7]
IDCOL. Project sponsor, technical review and eligible financing window	500,000 / 2,000,000 / 5,000,000	Project lending or R&D route; product-specific offer NOT FOUND; 4 / 9 / 18	0% / 3% / 10%	A; Solar terms are not passive-wind eligibility [31]
a2i / ICT innovation. Public service innovation and published eligibility	100,000 / 500,000 / 1,000,000	Grant, call-specific; 3 / 6 / 12	0% / 5% / 20%	A; Current matching call NOT FOUND [32]
GCF / accredited partner. Accredited entity or accepted project-specific route	0 / 0 / 2,000,000	No budgeted receipt; AE/PSAA project route; 9 / 18 / 36	0% / 1% / 5%	A; Direct startup allocation NOT FOUND [26]
BCCRF legacy. Do not assume legacy programme is open	0 / 0 / 0	Excluded from recommended cash plan; 0 / 0 / 0	0% / 0% / 0%	A; Current actionable call NOT FOUND [46]
Crowdfunding / preorders. Evidence, delivery plan and clear refund terms	100,000 / 300,000 / 700,000	A 5-15 percent platform/fulfilment reserve, refunds possible; 2 / 4 / 8	0% / 15% / 35%	A; Platform access and legal/FX terms NOT FOUND [39]

iDEA criteria and grant cap were checked in its official SOP; the SME route is via participating lenders. The official Bangladesh Bank listing confirms a green-factory refinance circular, but product-specific eligibility, borrower terms and credit approval remain unresolved. Startup Bangladesh FAQ terms are available in indexed text but full-page access failed. IDCOL solar terms must not be transferred to passive mechanical wind, and GCF support generally requires an accredited or accepted project-specific entity route. Current matching BCCRF/a2i/competition calls were NOT FOUND. [32, 40, 7, 43, 31, 26]

Pilot funding target	BDT	USD	Share	Basis
Founder cash	440,000	3,558	19.6%	[A] none committed
University contribution	400,000	3,235	17.9%	[A] none committed
CSR pilot	700,000	5,661	31.2%	[A] none committed
iDEA grant target	700,000	5,661	31.2%	[A] none committed

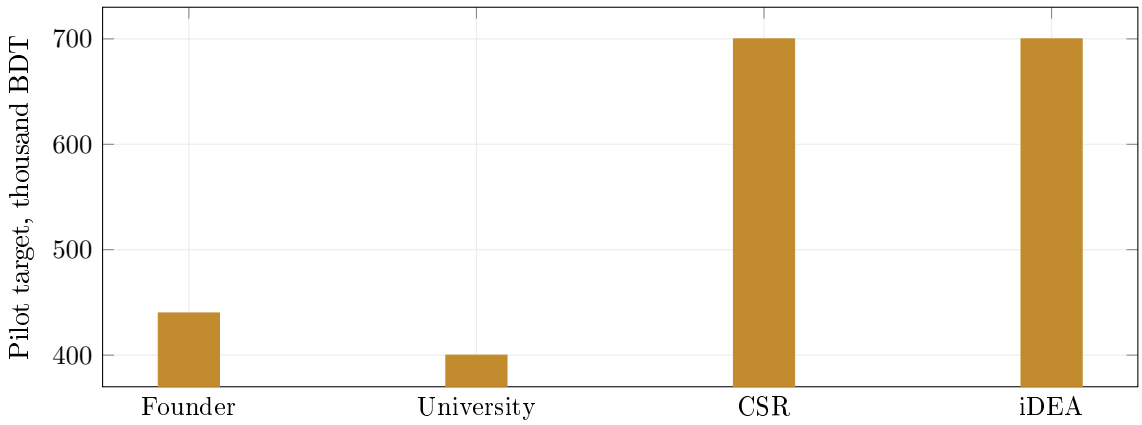


Figure 9: Recommended initial funding targets [A], not expected receipts. In-kind contributions offset the relevant budget item, and must not be counted again as unrestricted cash.

Release funds against gates: first confirm structural safety and a calibrated rotor/load curve; then demonstrate useful airflow or oxygen transfer in supervised conditions; then obtain written customer and supplier commitments. Pause scale spending if a gate fails. Debt should follow evidence of repayment capacity; a lower loan rate cannot turn a structurally negative product value proposition into a sound one.

11 Bangladesh sourcing, assembly and import exposure

Component group	Named source / assembly option	Lead time and landed-cost boundary	Basis
Patterns and filament	TechShopBD, HTE Bangladesh, Costareen, Electronics.Com.BD	Local stock indication; custom printed work requires quote	[S] observed offers [21, 22, 23, 24]
Gears and supports	ZerOneTech hobby sample; KHK specification catalogue; Motion/MISUMI bearing references	Production-equivalent matched bores and torque ratings require RFQ	[S/A] not interchangeable [27, 33, 9, 10, 11]
Fan and ducts	Walton distributor for conventional comparator; local sheet-metal workshop RFQ	No verified workshop quote for plenum/roof works	[S/A] [17]
Wet hardware / floats	NSC BD and Bangla Ltd. as commercial aerator benchmarks	Complete machines show materials; separate float/paddle supply RFQ needed	[S/A] [1, 4]
Custom shafts and frames	Candidate Nawabpur/Dholaikhal workshops; IUT facilities for supervised R&D	No named, vetted fabricator quote found; do not imply a purchase agreement	[A] sourcing route
Imports	Authorised dealer / freight forwarder; NBR classification	Freight, CD, VAT, advance taxes and broker charges depend on exact HS/SRO	[A] exact current duty NOT FOUND [14]

Assumed local lead time is 1 [A] / 3 [A] / 6 [A] weeks; import lead time 3 [A] / 6 [A] / 12 [A] weeks. A landed uplift reserve of 15% [A] / 35% [A] / 65% [A] is only a stress range. It is not a tariff ruling and is not applied again to BOM allowances already intended as domestic purchase budgets. Classified landed quotes are required before procurement.

Assembly acceptance should record rotor balance, shaft alignment, free play, axial retention, brake stop response, guard fasteners, roof water ingress and float leak/stability checks. Keep serialised service records and replaceable wear parts. Production tooling, inspection time and rejected parts are paid costs, not assumed free university support.

12 Future scope with costed evidence gates

Future item / test	Low / base / high BDT	Timing months	Source and decision boundary
Calibrated performance rig. Torque-speed, static-pressure/flow, DO transfer and brake-stop data	100,000 / 250,000 / 500,000	0–3	Core validated curves; no revenue booked; [A] budget [48, 3, 12, 19]
Structural and overspeed redesign. Guard retention, fatigue, anchor uplift and mechanical fail-safe	100,000 / 250,000 / 600,000	1–6	Independent review before occupied/public installation; [A] budget [48, 33]
Durable and recycled material pilot. UV, salt exposure, creep, leaks and recycled-lot coupons	50,000 / 150,000 / 350,000	2–6	Lower lifecycle cost must exceed rework cost; [A] budget [34, 2, 38]
Monitoring kit per air site. Calibrated CO2, anemometer, RPM and enclosure	7,000 / 18,000 / 40,000	3–6	Optional electrical load; no measured value premium; [A] budget [47, 12]
Monitoring kit per pond site. DO/temperature logging and reference calibration	12,000 / 35,000 / 80,000	3–6	Pre-dawn oxygen trend, sensor maintenance required; [A] budget [20, 3]
Fixed PV hybrid incremental kit. Matched module, motor, charge control and battery autonomy	20,000 / 38,000 / 70,000	6–12	Compare fixed PV against costly tracking; no autonomy guarantee; [A] budget [39, 48]
Regional service and modular fleet trial. Multiple independently metered assets and response-time records	100,000 / 300,000 / 700,000	9–18	Separate additive power from correlated wind availability; [A] budget [19, 48]
Grain grinding feasibility study. Torque requirement, gearing, food-contact and guarding tests	50,000 / 150,000 / 300,000	12–24	Deferred; not a current business model; [A] budget [36, 33]

Each citation supports a method or engineering principle, not an asserted benefit percentage. Spiral-turbine literature concerns other geometries and operating points; no published Cp is copied into a claim about this prototype. Clean-water oxygen-transfer standards and FAO oxygen budgets support testing, while clinic/WHO material supports measurement and intake design. Costs are independent planning alternatives and can overlap the launch test budget; they are not all added to the financial projection. [41, 42, 3, 19, 12, 47]

12.1 Roadmap and staged expansion

The proposed sequence below requires evidence before each expansion step. Test infrastructure, an independent safety review and durability data precede optional monitoring and fleet growth. Costs and dates remain assumptions; an unsuccessful test should delay the next stage rather than trigger automatic deployment.

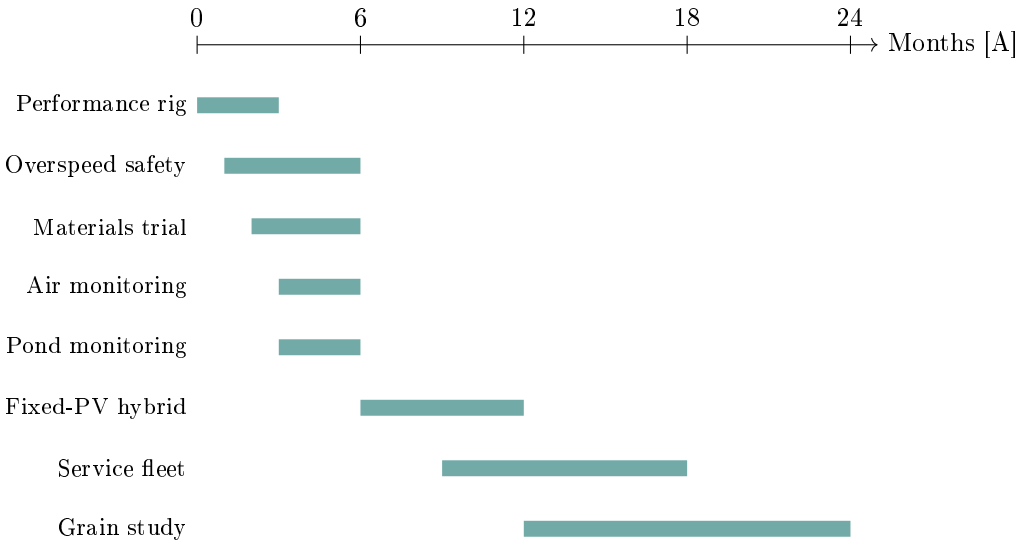


Figure 10: Future-scope sequence [A]. Advancement depends on evidence gates, not elapsed time alone.

Solar tracking shown in the site adds moving parts; a fixed-panel pilot should establish whether assistance value justifies the extra cost. More pond units add potential transfer but remain exposed to shared lulls. Grain grinding appears only in a submission sketch and remains deferred until torque, food-contact, guarding and economics are tested. No future-feature revenue is included in NPV.

13 Environmental screening and end of life

The mechanical core uses no operating electricity, but it has embodied materials, fabrication, logistics, lubricant and replacement impacts. A transparent screening calculation is preferable to a zero-carbon claim. The DoE published factor is 0.62 [S] kg CO2/kWh for the stated older fiscal periods, published in its January notice. This historical average factor is not a prediction of future marginal displacement or a verified carbon-credit methodology. [28]

Gross avoided emissions equal displaced grid kWh times the grid factor. Net screening benefit subtracts assumed embodied emissions. Foreground masses are 15 [A] kg steel, 4 [A] kg aluminium and 3 [A] kg polymer for A, or 8 [A] kg for B. Assumed factors are 2.0 [A], 8.0 [A] and 3.0 [A] kgCO2e/kg respectively. These are screening placeholders, not supplier EPDs; bounds span half to double the resulting embodied estimate. Hybrid PV/battery/electronics receive an additional lifecycle reserve, not a complete LCA.

Case	Embodied low / base / high kgCO2e	Gross avoided over horizon kgCO2	Net screening kgCO2e	Carbon payback years / basis
A	60 / 121 / 242	703.7	582.7	1.7; [D], not verified LCA
H	210 / 421 / 842	1,269.5	848.5	3.3; [D], not verified LCA
B	68 / 136 / 272	85.9	−50.1	15.8; [D], not verified LCA

Negative net screening values are retained. Metal recovery is plausible in principle, but no supplier take-back or local recycling recovery rate has been verified. The recovery target in the model is a design target only. Keep material labels, demountable joints, drainable housings and captured lubricants. Recover printed fragments and damaged floats; do not permit pond disposal. Batteries require a dedicated recycler/take-back arrangement.

Noise, structure-borne vibration, wildlife strike/entanglement, corrosion products and aquatic disturbance are NOT MEASURED. Small size and a spiral shape do not prove silence or ecological safety. Use guards and setbacks, survey the site, log vibration/noise at occupied locations, and inspect water impacts. Increasing airflow can introduce outdoor PM and moisture; filtration or damper changes can consume the small pressure budget. [48, 34, 20]

14 Risks, limitations and decision gates

Risk	Evidence consequence	Action / owner	Basis
Unknown loaded startup and torque	Core performance cannot be certified	Engineering team: measured torque-speed/load curve before commercial claims	NOT FOUND
Wind lulls and site turbulence	Neither time availability nor outage coincidence verified	Independent site assessor: hub-height logging and demand/wind joint analysis	[S/A] [48, 49]
Storm and structural safety	Code rotates in storms; no hardware cut-out proof	Qualified engineer: fatigue, anchorage, guarding and fail-safe test	NOT FOUND
Critical clinic / stocked pond liability	Energy example is not safe service delivery	Site owner and expert: define service demand, backup and alarm response	[S/A] [12, 18]
Custom quotations and warranty	BOM bands may miss costs	Procurement: equivalent-spec written RFQs and pilot failure records	NOT FOUND
Demand and tariff acceptance	Proposed revenue may not be realisable	Commercial team: paid pilot, competing offers and customer interviews	NOT FOUND
Tax, customs and insurance	Scenario reserves may differ from actual cash outlays	Accountant/broker/insurer: current entity and classification-specific advice	NOT FOUND
Future-source access and age	Indexed or dated sources limit confidence	Maintain date/access flags and replace with primary full text	[S] source register

Discount-rate sensitivities are not a substitute for modelling correlated failure, supplier escalation or market rejection. The project has no verified production CAD, independent lifetime tests, customer orders or guaranteed grant. All recommendations remain conditional on the explicitly recorded assumptions. A failure to meet an evidence score threshold is reported rather than repaired by raising a self-assigned score.

15 Conclusion

ZeroWatt Wind is a credible student mechanical concept with an honest poster boundary, but current evidence does not support commercial-scale claims. The base project NPV is −11,181,078 [D] BDT and its annual effective IRR is −47.7% [D]; payback is not reached. The appropriate next investment is a gated measurement and durability pilot. Product redesign, substantially lower cost or independently demonstrated non-energy value would be required before broad rollout. No health, fish-yield, resilience or carbon premium should be promised in advance of evidence.

16 Assumptions and unresolved-evidence register

Input	Low	Base	High	Basis / reason
FX BDT/USD	123.650	123.650	123.650	S: BB WAR 2026-10-19; not guaranteed transaction rate
Site mean wind m/s	2.500	3.500	4.500	A: No hub-height time series; utility-height speeds not transferred to roof
Mechanical availability	0.800	0.900	0.950	A: Includes maintenance/curtailment; prototype uptime unmeasured
Wind distribution k	1.500	2.000	2.500	A: Only k=2 used in central scenarios; k bounds flagged for later data fitting
Price multiplier	0.850	1.000	1.100	A: No willingness-to-pay survey
Adoption multiplier	0.500	1.000	1.800	A: No orders or customer funnel
Production cost multiplier	0.900	1.000	1.250	A: No comparable RFQs for custom production BOM
Annual price/cost escalation	0.020	0.040	0.080	A: Common nominal escalation; no margin inflation benefit
Tax reserve fraction	0.200	0.250	0.300	A: Entity-specific statutory taxes NOT FOUND; no tax advice
Yearly churn	0.040	0.080	0.150	A: No retention observations
Uncollected revenue	0.015	0.030	0.080	A: No payment histories
Lease BDT/month	1,000.000	1,500.000	2,000.000	A: Supplemental access tariff; not an energy-saving price quote
Pay-per-use BDT/h	20.000	30.000	40.000	A: Metered delivery and collection untested
Paid hours scheduled/month	60.000	120.000	180.000	A: Actual billable hours multiplied by energy-service fraction
Installation fee BDT	2,000.000	3,000.000	5,000.000	A: Unbundled sold units only
AMC fee BDT/year	1,000.000	1,800.000	3,000.000	A: After warranty; separate sold units only
Grid benchmark BDT/kWh	9.050	15.360	20.000	A/S: 2026 BERC LT-D1 and LT-E S; future tariff high A; VAT/fixed costs cancel for existing connection
Electric field SAE kg/kWh	0.800	1.200	1.800	A: FAO and supplier standard ratings are not field delivery
Wind field oxygen correction	0.300	0.500	0.700	A: No alpha/beta or site water test
Wind standard SAE kg/kWh	0.400	0.800	1.200	A: Site default base .8, prototype test NOT FOUND
Diesel BDT/litre	90.000	110.000	140.000	A: No verified current pump quote
Generator litre/kWh	0.300	0.400	0.600	A: Part-load sharing; own set at tiny load could be much worse
Embodied footprint multiplier	0.500	1.000	2.000	A: Foreground masses and factors not supplier EPDs
Import landed uplift	0.150	0.350	0.650	A: Freight duties and broker reserve; not a statutory tariff
Local lead time weeks	1.000	3.000	6.000	A: Custom RFQ not received
Import lead time weeks	3.000	6.000	12.000	A: No shipment quote
Discount rate nominal	0.120	0.180	0.250	A: Historical 10 percent BB policy benchmark plus project risk; not observed WACC
AMC uptake	0.200	0.400	0.600	A: No service-contract demand study
Customer mechanical annual maintenance share	0.020	0.040	0.080	A: Service visits can dominate low output
Hybrid avoided energy fraction addition	0.250	0.450	0.600	A: Applied to base wind fraction, capped .85; dispatch model NOT FOUND

Base assumptions are used as written, not selected to produce a positive outcome. Scenario input combinations are neither statistical confidence intervals nor independent probabilities. Ranges without direct evidence are labelled [A] and should be replaced with measured distributions or written quotations before investment.

NOT FOUND / low-confidence item	**Required evidence**
Measured core performance	Starting wind speed, loaded torque/RPM, Cp, drivetrain loss, airflow/static pressure, field and standard oxygen transfer.
Durability and safety	Production stress/fatigue design, safe storm shutdown, anchor/mooring loads, float reserve buoyancy, weathering and service life.
Hub-height wind	Bangladesh rooftop/pond distribution, fraction meeting demand, longest lulls and coincidence with outages/pre-dawn DO need.
Commercial proof	Current baseline business models/prices in supplied materials, purchase commitments, willingness to pay, retention, collections and market funnel.
Component quotations	Three independent comparable production price points for every custom rotor/frame/gear/guard/float/duct and warranted hybrid subsystem.
Supplier execution	Verified custom Bangladesh fabricator RFQs, production lead times, landed freight and current HS/SRO tax incidence.
Operating/legal costs	Actual insurance, entity-specific tax, minimum/turnover taxes, VAT recoverability, warranty incidence and fleet loss rates.
Funding	Programme-specific eligibility, current matching BCCRF/a2i/competition calls, grant success probabilities, committed awards, investor terms and crowdfunding access.
Attributable benefits	Local avoided health cost/time, cognitive productivity gain, fish mortality/yield improvement and monetary resilience value.
Environmental proof	Supplier EPDs, material masses, recycler recovery rates, audited lifecycle inventory, noise/vibration and wildlife/aquatic impact.
Future scope	Measured tracking gain, safe grain-grinding output, validated regional fleet operation and service economics.
Blocked source verification	Several official/vendor full texts were blocked or index-only; those records carry explicit access limitations.
Gauntlet sufficiency	Evidence-dependent tasks and the financial model may remain below the requested evidence threshold after the maximum rounds; no score is fabricated.

17 Gauntlet log summary and reproducibility

The repository preserves a draft, five task-specific objections and successive source/model repairs for each requested task, plus finance. This is one analyst self-review using investor, judge and fact-checker perspectives; it is not an independent expert review. Evidence sufficiency, arithmetic, realism and completeness are scored separately. Transparent assumptions do not turn an evidence gap into a sourced measurement.

Task	Passes	Scores E / N / R / C	Remaining limitation / Basis
01	3	8 / 9 / 9 / 9	Requested evidence threshold met for bounded description; physical validation still separate. [A] self-assessment
02	6	6 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
03	6	7 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
04	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
05	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
06	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
07	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
08	6	5 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
09	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
10	6	7 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
11	6	7 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
12	3	9 / 10 / 9 / 9	Requested evidence threshold met for bounded description; physical validation still separate. [A] self-assessment
13	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
14	6	7 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
15	6	7 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
16	6	6 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment

Task	Passes	Scores E / N / R / C	Remaining limitation / Basis
17	6	7 / 9 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
18	3	8 / 9 / 9 / 9	Requested evidence threshold met for bounded description; physical validation still separate. [A] self-assessment
19	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
20	6	6 / 10 / 9 / 9	NOT FOUND: task-specific field data, comparable quotes or commercial proof; Evidence <8 after maximum rounds. [A] self-assessment
Finance	6	6 / 10 / 9 / 9	Unvalidated demand, WTP, resource and production quote inputs; negative base economics retained. [A] self-assessment

Reproduction order: run the site audit with Node, run the standard-library Python financial model, then run the report generator. Monetary calculations originate in the model exports; all display rounding and embedded chart data are registered. The standalone TeX uses no external image, bibliography or data file. Source IDs in every citation are checked against the opened-source register. Full monthly/annual tables, assumptions, price observations, BOM, customer comparison, sensitivity and self-check outputs accompany the report.

The financial run passes 51 [D] accounting, physical-bound and root-residual self-checks. Compilation and visual QA are documented separately in the repository; arithmetic tests do not validate commercial assumptions.

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